## JC 2 Preliminary Examination Paper 3 Answer key

**1(a) Describe how the bee gene could have been inserted into the plasmid.**

- Bee gene is excised/isolated/ cut/ spliced from genome using restriction enzymes that gives sticky ends
- The same restriction enzyme is used to cut/ splice the plasmid
- The bee gene and the cut/ spliced plasmid is mixed together
- The sticky ends of the excised/ isolated bee gene and plasmid will anneal / join via complementary base pairing
- DNA ligase will catalyse the formation of a phosphodiester bond to seal the bee gene to the plasmid. Max [3]

**(b) These plasmids were injected into the fertilised eggs of mosquitoes, rather than into adult mosquitoes. Explain why.**

- Injecting plasmid into fertilised egg will ensure that the bee gene that is expressed in all cells of the mosquito
- So the gene will be passed on to future generations.
- in adult mosquitoes, the bee gene will only be found in somatic cells and these cells will not be passed on to the next generation. max [2]

**(c) Within the plasmid contained a genetic marker, a gene encoding a fluorescent protein that produces green light. Explain why this marker gene was used in this experiment.**

- The florescent gene will floresence in the dark
- Will allow for easy identification of mosquitoes that have taken up the bee gene. Max[2]

**(d) Many years ago humans accidentally introduced a species of mosquito to some Hawaiian islands. This species of mosquito carries a type of malaria that only infects birds. Some of these birds are rare. It has been suggested that the control of malaria using genetically-engineered mosquitoes with the bee gene should be tested on these Hawaiian islands. Suggest one advantages of using this approach.**

- Could save birds from malaria, so reversing harm done by humans;
- Islands isolated, so fewer problems if trial goes wrong;
- If malaria parasite becomes resistant to bee protein, resistance not in human
- malaria strain;
- Resistance will spread through the mosquito population; any 1 point [1]

**(e) Explain why during each cycle,**

**(i) the DNA is heated to 95°C.**

- at 95°C the hydrogen bonds holding the double stranded DNA together will be broken,
- making the DNA single stranded_ max [1]

**(ii) then reaction mixture is cooled to 40°C.**

- will allow the re-annealing of the DNA strands via complementary base pairing.
- But RNA primers are in excess, thus primers will anneal to the single stranded DNA instead max [1]

**(iii) and the temperature is again raised, but to 70°C.**

- Optimum working temperature for Taq polymerase
- Nucleotides are added to ends of primers and new strand of DNA will be synthesized max [1]

**(f) In polymerase chain reaction, suggest why DNA synthesis takes place in opposite directions on the two separated strands of the double helix.**

- the strands are anti-parallel
- Taq DNA polymerase only functions in the 5'nto 3' direction max [2]

**(g) State one advantage and one disadvantage of using *Taq* DNA polymerase.**

advantage: thermostable, allows automation of PCR (1m)
disadvantage: No 3' to 5' proofreading activity (1m) max [2]

**[Total: 15]**

**2(a) Explain briefly:**

**(i) How the DNA fragments are separated by gel electrophoresis.**

- Molecular size of DNA
- Agarose / polyacrylamdie is porous matrix/ has pores
- Smaller molecules move through the gel faster, larger molecules slower
- DNA negatively charged and will move towards the positive pole / anode [2]

**(ii) The basis behind RFLP.**

- Different alleles/ short tandem repeats (STR) have differences / alterations to nucleotide sequences in alleles/ STRs
- thus alter restriction sites within alleles/ STRs
- resulting is varying lengths of DNA after restriction enzyme reaction max [2]

**(b) A marker DNA should be loaded onto the gel together with these DNA samples. State the role of a marker DNA.**

- Marker DNA is of known sizes
- Serve as a reference to estimate the size of DNA fragments separated by gel electrophoresis
- Through comparison of relative position on the gel to marker DNA max [1]

**(c) A southern blot was prepared and probes, specific for certain RFLP markers in human beings, were added.**

**a. How do probes detect the specific RFLP markers present on the different individuals?**

- Probes are single stranded and labelled with radioactivitiy
- anneal to the single stranded RFLPs via complementary base pairing
- autoradiography performed and
- probe will show up as a dark band max [2]

**(ii) Give an example of an RFLP marker used in forensic science or paternity testing.**

Tandemly repetitive DNA sequences [1]

**(d) Which pair of parents is likely to be the parents of the dead soldier? Explain your choice.**

- parents C and D
- Offspring should inherit half of genome of each parent
- Bands 2,3,4,6 and 10 is from parent C while 1,5,7,8 and 9 from parent D,
- Thus restrictions sites must be from C and D max [2]

(e) State

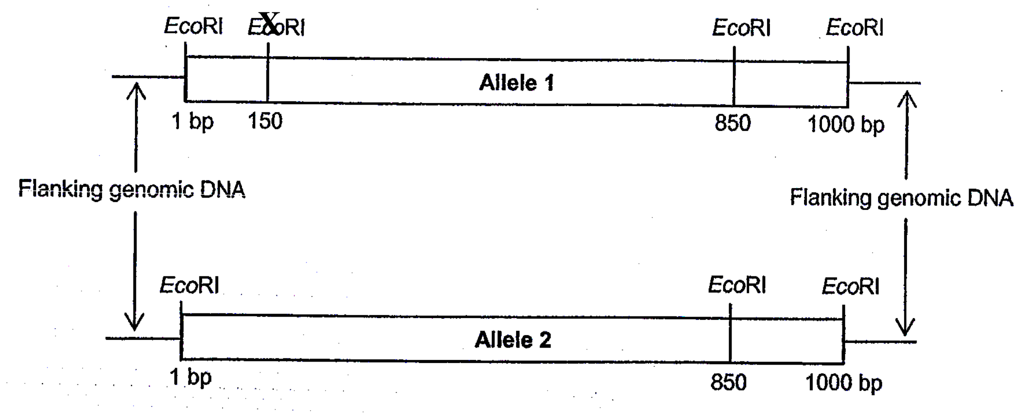
(i) **the number of bands, seen on the gel, from this length of DNA produced from the pair of homologous chromosomes.**

5 bands [1]

(ii) **The different sizes of these fragments.**

3, 15, 5, 10, 20kb [1]

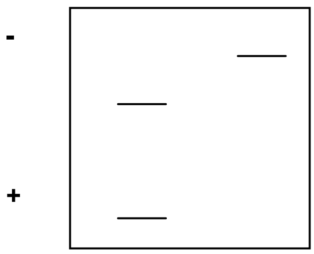
The diagram below shows the restriction site present in two alleles that are 1000bp long.




(f) (i) **On Allele 1, indicate the position of a probe that you would design to distinguish the two alleles.** [1]

spot with X

(ii) **In the gel below, draw the resulting autoradiogram you would expect to see based on the probe you have drawn.** [2]




**[Total: 15]**

**3(a) Describe what is meant by somatic cell gene therapy.** **[1]**

- Refers to introduction of gene(s) into cells that are not destined to become gametes to cure a genetic disease (cure not extended to future generations)

**(b) Explain what is meant by an X-linked disease, and suggest how such diseases are first noticed in populations.** **[2]**

- Genetic disease, whereby the allele coding for the condition is carried on the X-chromosome;
- Transmission usually from mother to son as males inherit one copy of X chromosome from mothers / affected mothers produce affected sons but not unaffected daughters / males express the trait;

**(c) Describe two properties of retroviruses that make them suitable as vectors for gene therapy.** **[2]**

- Possess integrase which can integrate the target gene into the genome of patient's cells
- Ability to infect rapidly dividing cells and thus is able to be used on cancer cells.

**(d) Explain why patients treated with retroviruses are at significant risks of contracting leukaemia. [2]**

- Integration of the targeted gene might be on site away from the target site.
- Might activate certain proto-oncogene to oncogene or deactivate tumour suppressor genes
- If a blood stem cell is affected, it might lead to excessive proliferation of a particular type of blood cell (for eg white blood cells), leading to leukaemia.

**(e) Explain how DNA transferred by adenoviruses could help patients even though the DNA is not incorporated into the genome. [2]**

- The DNA is inserted into cell cytoplasm and transcribed / translated into proteins
- Without entering the nucleus or integration into genome;
- Since a functional protein / enzyme can now be produced by the target cells, the patients might be temporarily cured.

**(f) Elaborate on a feature of the liposome that makes it suitable as a vector for gene transfer. [2]**

- Similar structure with plasma membrane (phospholipid bilayer)
- Enables the liposome to fuse with the plasma membrane of target cells to transfer gene

**(g) Based on the information provided, describe how liposomes could be used to treat named respiratory disease. [3]**

- Named respiratory disease like cystic fibrosis
- Due to a non-functional CFTR (cystic fibrosis transmembrane conductance regulator) protein regulating chloride transport
- Liposomes carrying the gene coding for functional CFTR could be inhaled to transfer the gene to cells of the lung
- If the gene is taken up by the cell, functional proteins could be produced.

**(h) Based on the understanding that every current vector for gene therapy has its strengths and weaknesses, suggest how future vectors for gene therapy might be like. [1]**

- Fusion or hybrid gene therapy involving the use of several vectors in combination
- Viral particles carrying the gene for transfer could be enclosed within liposomes.

**[Total: 15]**

4 (a) **Describe how genetic engineering has helped in improving the quality and yield of named crop plants and animals to solve the demand for food in the world [8]**

Overall description of genetic engineering

- the transfer of genetic material from one organism and inserting it into the permanent genetic code of another;
- Possibility of engineering numerous novel creations because of universality of genetic code;
- Creation of recombinant / genetically modified organism (definition);

Quality (allow 2 examples (2m each), 1 plant and 1 animal)

- Vitamin / nutrient enhanced food (beta carotene-enhanced rice);
- Rice enhanced with iron content;
- Increased quality of meat yield (less fat and more proportion of meat)
- Milk producing vaccines for diseases.

Yield (allow 2 examples (2m each), 1 plant and 1 animal)

- Insect-resistant plants; (named example such as in BT plants);
- rate of growth (maturity) of cows / chicken / fish due to alteration of growth hormone expression;
- Herbicide tolerance
- Producers and consumers of food tend to benefit from increased production or decreased mortality of crops / animals.

(b) **Discuss the potential risks of genetically modifying organisms. [6]**

Personal risk

- Food safety issues (potential problems)
    - Toxicity
    - Allergenicity
    - Antibiotic resistance
    - Immune suppression
    - Cancer

Ecological risks (agriculture)

- Effects on wildlife
    - Competing with other organisms
    - Poisoning of other wildlife
- Effects on growth of organisms that are not targeted (weeds and pests)

- Bio-terrorism / abuse

- For cloned organisms, might be susceptible to same diseases

(examples may be awarded marks accordingly)

(c) **Explain the normal functions of a named type of adult stem cell. [6]**

- Named example such as haematopoietic stem cell
- Ability to differentiate into cells found in the blood (red blood cells, white blood cells and platelets)

Features of adult stem cells

- Multipotency and definition as ability to generate cells of a particular tissue type but not others / not whole organism / individual.
- Self-renewal as the ability to generate a new cell via mitosis
- where one cell remains as the adult stem cell while the other differentiates to become a specialised cell
- Relatively undifferentiated (ability to specialise)

Functions (linking properties to functions)

- Growth of organism
- Repair of damaged cells / tissue
- Stem cells sent to injury sites for cell repair (mechanism similar to growth)